

JAVA PROGRAMMING ASSIGNMENT

NAME = ROHIT KHANDELWAL
SECTION = G
ROLL NUMBER = 25EPTCX050
SUBJECT = JAVA PROGRAMMING
BATCH = G3

QUESTION 1:-

<https://github.com/2025pietcdrohit050-ux/java-assignment/blob/main/que%201>

JAVA CODE:-

```
import java.io.*;

class StudentGradeCalculator {

    public static void main(String[] args) throws IOException {

        BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

        System.out.print("Enter marks of Subject 1: ");
        int sub1 = Integer.parseInt(br.readLine());

        System.out.print("Enter marks of Subject 2: ");
        int sub2 = Integer.parseInt(br.readLine());

        System.out.print("Enter marks of Subject 3: ");
        int sub3 = Integer.parseInt(br.readLine());

        System.out.print("Enter marks of Subject 4: ");
        int sub4 = Integer.parseInt(br.readLine());
```

```
System.out.print("Enter marks of Subject 5: ");
int sub5 = Integer.parseInt(br.readLine());

int total = sub1 + sub2 + sub3 + sub4 + sub5;
double percentage = (double) total / 5;

String grade;

if (percentage >= 90) {
    grade = "A+";
} else if (percentage >= 80) {
    grade = "A";
} else if (percentage >= 70) {
    grade = "B";
} else if (percentage >= 60) {
    grade = "C";
} else if (percentage >= 50) {
    grade = "D";
} else {
    grade = "Fail";
}

System.out.println("\n----- Result -----");
System.out.println("Total Marks = " + total);
System.out.println("Percentage = " + percentage + "%");
System.out.println("Grade = " + grade);

if (percentage >= 50) {
    System.out.println("Status = Passed");
} else {
```

```
System.out.println("Status = Failed");  
}  
}  
}
```

QUESTION 2:-

<https://github.com/2025pietcdrohit050-ux/java-assignment/blob/main/que%202>

JAVA CODE:-

```
import java.io.*;  
  
class ElectricityBill {  
    public static void main(String[] args) throws IOException {  
  
        BufferedReader br = new BufferedReader(new  
InputStreamReader(System.in));  
  
        System.out.print("Enter Consumer Name: ");  
        String name = br.readLine();  
  
        System.out.print("Enter Consumer Number: ");  
        String number = br.readLine();  
  
        System.out.print("Enter Units Consumed: ");  
        int units = Integer.parseInt(br.readLine());  
  
        double bill = 0;  
  
        if (units <= 100) {  
            bill = units * 3;  
        } else {
```

```

        if (units <= 200) {
            bill = (100 * 3) + (units - 100) * 5;
        } else {
            if (units <= 500) {
                bill = (100 * 3) + (100 * 5) + (units - 200) * 7;
            } else {
                bill = (100 * 3) + (100 * 5) + (300 * 7) + (units - 500)
* 10;
            }
        }
    }

    if (bill > 5000) {
        bill = bill + (bill * 5 / 100);
    }

    System.out.println("\n----- Consumer Details -----");
    System.out.println("Consumer Name    : " + name);
    System.out.println("Consumer Number : " + number);
    System.out.println("Units Consumed  : " + units);
    System.out.println("Total Bill      : Rs. " + bill);
}
}

```

QUESTION 3:-

<https://github.com/2025pietcdrohit050-ux/java-assignment/blob/main/que%203>

```

import java.io.*;

class NumberOperator{

    public static void main(String[] args) throws IOException {

        BufferedReader br = new BufferedReader(new
InputStreamReader(System.in));
    }
}

```

```
int num, c=1;
System.out.print("enter the number:");
num = Integer.parseInt(br.readLine());
while(c!=6){
System.out.println("enter the number to do following actions:");
System.out.println("1. Check whether the number is Prime");
System.out.println("2. Check whether it is Palindrome");
System.out.println("3. Check whether it is Armstrong");
System.out.println("4. Find the Sum of Digits");
System.out.println("5. Reverse the Number");
System.out.println("6. Exit");
c = Integer.parseInt(br.readLine());
switch(c) {
    case 1:
        int prime = 1;
        for(int i=2; i<num; i++){
            if(num%i==0){
                System.out.println("its not a prime");
                prime = -1;
                break;
            }
        }
        if(prime==1) System.out.println("it is a prime");
        break;
    case 2:
        int copy=num;
        int rev=0;
        while(copy!=0){
            rev= (rev*10) + (copy%10);
            copy/=10;
```

```

}
if(num==rev) System.out.println("given number is a Palindrome");
else System.out.println("given number is not a Palindrome");
break;
case 3:
int copy1=num;
int n=0;
while(copy1!=0){
    n++;
    copy1/=10;
}
copy1=num;
double arms=0;
while(copy1!=0){
    int tamp= copy1%10;
    arms = arms + Math.pow(tamp, n);
    copy1/=10;
}
if(num==arms) System.out.println("given number is a Armstrong");
else System.out.println("given number is not a Armstrong");
break;
case 4:
int copy2=num;
int sum=0;
while(copy2!=0){
    int tamp = copy2%10;
    sum = sum + tamp;
    copy2/=10;
}
System.out.println("the sum of Digits is:" + sum);

```



```

System.out.println("enter 10 numbers");
for(int i=0; i<10; i++) arr[i]= Integer.parseInt(br.readLine());
int[] rev = new int[10];
int low=0, high=0, even=0, odd=0, key, found=-1;
double avr=0;
System.out.println("enter key value to search:");
key= Integer.parseInt(br.readLine());
for(int i=0; i<10; i++){
    if(arr[i]%2==0) even++;
    else odd++;
    if(arr[i] > arr[high]) high = i;
    if(arr[i] < arr[low]) low = i;
    avr+=arr[i];
    rev[9-i]=arr[i];
    if(arr[i]==key) found=i;
}
avr/=10;
System.out.println("highest number:" + arr[high]);
System.out.println("lowest number:" + arr[low]);
System.out.println("total even numbers:" + even);
System.out.println("total odd numbers:" + odd);
System.out.println("avreage:" + avr);
System.out.println("found " + key + " at index " + found);
System.out.print("array in reverse order:");
for(int i=0; i<10; i++) System.out.print( rev[i] + " ");
}
}

```

QUESTION 5:-

<https://github.com/2025pietcdrohit050-ux/java-assignment/blob/main/que%205>

JAVA CODE:-

```
import java.io.*;
```

```
class Employee {
```

```
    int emp_id;
```

```
    String emp_name;
```

```
    int BS;
```

```
    double DA, HRA, PF, GS, net;
```

```
    BufferedReader br = new BufferedReader(  
        new InputStreamReader(System.in));
```

```
    void calculate() {
```

```
        DA = 0.4 * BS;
```

```
        HRA = 0.2 * BS;
```

```
        PF = 0.12 * BS;
```

```
        GS = BS + DA + HRA;
```

```
        net = GS - PF;
```

```
    }
```

```
    void get() throws IOException {
```

```
        System.out.println("Enter the Employee id:");
```

```
        emp_id = Integer.parseInt(br.readLine());
```

```
        System.out.println("Enter the Employee name:");
```

```
        emp_name = br.readLine();
```

```
        System.out.println("Enter the Employee basic salary:");
```

```

        BS = Integer.parseInt(br.readLine());

        calculate();
    }

    void show() {

        System.out.println(emp_id + " | " + emp_name + " | "
            + BS + " | " + DA + " | " + HRA + " | "
            + PF + " | " + GS + " | " + net + " |");
    }
}

class Main extends Employee {

    public static void main(String[] args) throws IOException {

        BufferedReader b = new BufferedReader(
            new InputStreamReader(System.in));

        int n;

        System.out.println("How many Employee details you want to
input:");
        n = Integer.parseInt(b.readLine());

        Employee[] emp = new Employee[n];

        for (int i = 0; i < n; i++) {

            System.out.println("Enter the following details of "

```

```

        + (i + 1) + " Employee:");

    emp[i] = new Employee();
    emp[i].get();
}

System.out.println(
    "ID | NAME | BASIC SALARY | DA | HRA | PF | GS | NET
SALARY");

System.out.println(
    "-----
---");

for (int i = 0; i < n; i++)
    emp[i].show();

System.out.println(
    "-----
---");

int high = 0, low = 0, average = 0;

for (int i = 0; i < n; i++) {

    if (emp[i].BS > emp[high].BS)
        high = i;

    if (emp[i].BS < emp[low].BS)
        low = i;
}

```

```
        average = average + emp[i].BS;
    }

    average = average / n;

    System.out.println("Highest Paying Employee: "
        + emp[high].emp_name + " "
        + emp[high].emp_id + " "
        + emp[high].BS);

    System.out.println("Lowest Paying Employee: "
        + emp[low].emp_name + " "
        + emp[low].emp_id + " "
        + emp[low].BS);

    System.out.println("Average Salary: " + average);
}
}
```